BANK FOR INTERNATIONAL SETTLEMENTS

Monetary facts and monetary policy trade-offs: New evidence

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The views expressed are those of the authors and do not necessarily reflect those of the BIS.

Introduction

- Broader question of monetary policy's impact on and role in maintaining financial stability

- Popular way to think about how monetary policy affects financial stability (e.g. Svensson (2015)):
  - Monetary policy → Credit → Financial stability

- Links calibrated based on historical evidence (e.g. Schularick and Taylor's (2012) estimates of the credit-crisis link)

- Point of this presentation: The relevant links may have changed over time in the wake of changes in the monetary and financial regimes (price stability and financial liberalisation)

Outline

- Two parts
  - Monetary facts revisited (Gertler and Hofmann (2016))
    - Assesses how monetary facts, including the credit-crisis nexus, have evolved over the post-war period
  - Monetary policy transmission and trade-offs in the United States: Old and new (Hofmann and Peersman (2017))
    - Assesses how U.S. transmission and trade-offs have evolved over the post-war period

Monetary facts revisited

Pavel Gertler and Boris Hofmann

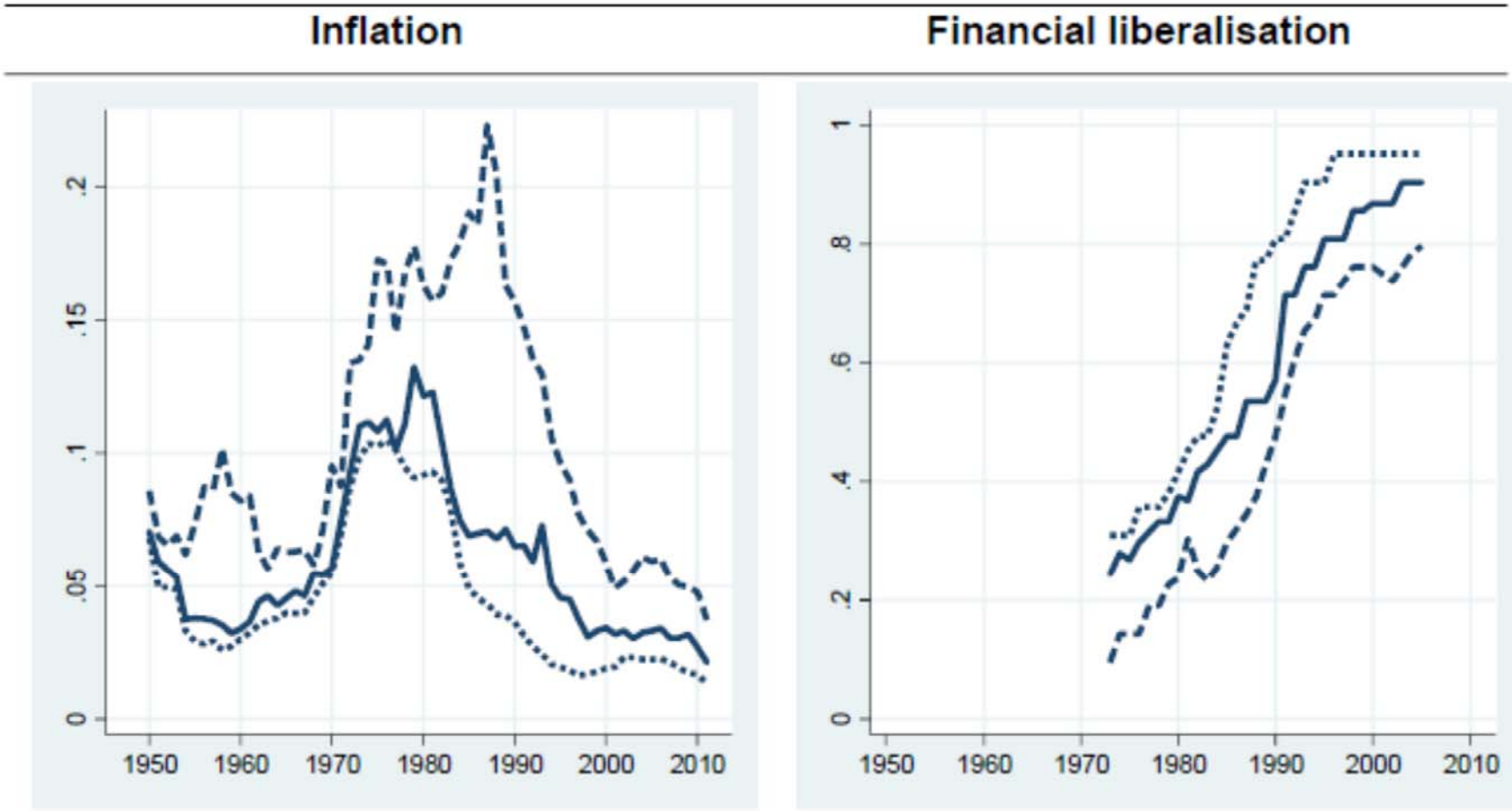
BIS WP No 566 (June 2016)

Two monetary facts

- One-to-one long-run link between money growth and inflation
  - e.g. McCandless and Weber (1995), Lucas (1996), Lucas (1980), Assenmacher-Wesche and Gerlach (2007)
  -
- Credit growth predicts financial crises
  - Borio and Lowe (2002a,b, 2004), Eichengreen and Mitchener (2003), Schularick and Taylor (2012)

- Monetary facts may however change over time and differ across countries due to changes/differences in the monetary and financial regime

Disinflation and financial liberalisation



Monetary and financial regime and monetary facts

- Velocity shifts due to financial innovation and disinflation may drive a wedge between money growth and inflation

- Financial liberalisation has increased frequency of credit boom-bust cycles; in low inflation regimes unsustainable booms may show up primarily in financial imbalances

- Low inflation and financial liberalisation may have weakened the money growth-inflation nexus and strengthen the credit-crisis nexus

- Evidence of weakening money growth-inflation link, no evidence so far of changes in credit-crisis link

Scope of this paper

- Assess changes in monetary facts over post-war period more systematically
- Annual data for 46 major AEs and EMEs over the period 1950-2011
- Data: GDP deflator, real GDP, broad money, bank credit, financial crisis indicator (CPI, narrow money, real exchange rate in robustness checks)
- Analysis based on standard single-equation panel econometric techniques
- Aim is to identify potentially relevant changes in correlations (monetary facts)

Econometric approach

- Money-inflation nexus: Pooled Mean Group (PMG) estimator taking into account McCallum/Nelson (2011) critique

$$\Delta^2 P_{it} = \theta_{0i} + \phi\left(\Delta P_{it-1} - \theta_1 \Delta M^E_{it-j}\right) + \sum_{j=0}^{4} \delta_j \Delta^2 M^E_{it-j} + \varepsilon_{it}$$

- Credit-crisis nexus: Panel probit following Schularick and Taylor (2012)

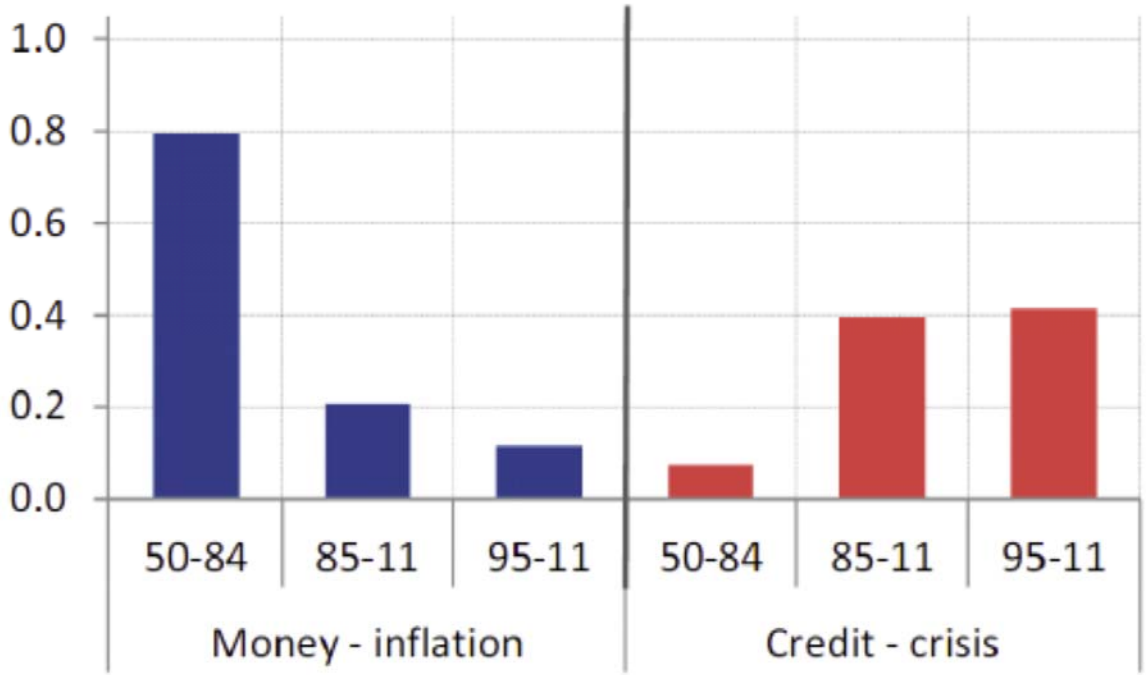
$$p_{it} = \lambda_0 + \sum_{j=1}^{5} \lambda_{1j} \Delta C^{RE}_{it-j} + \eta_{it}$$

Baseline results

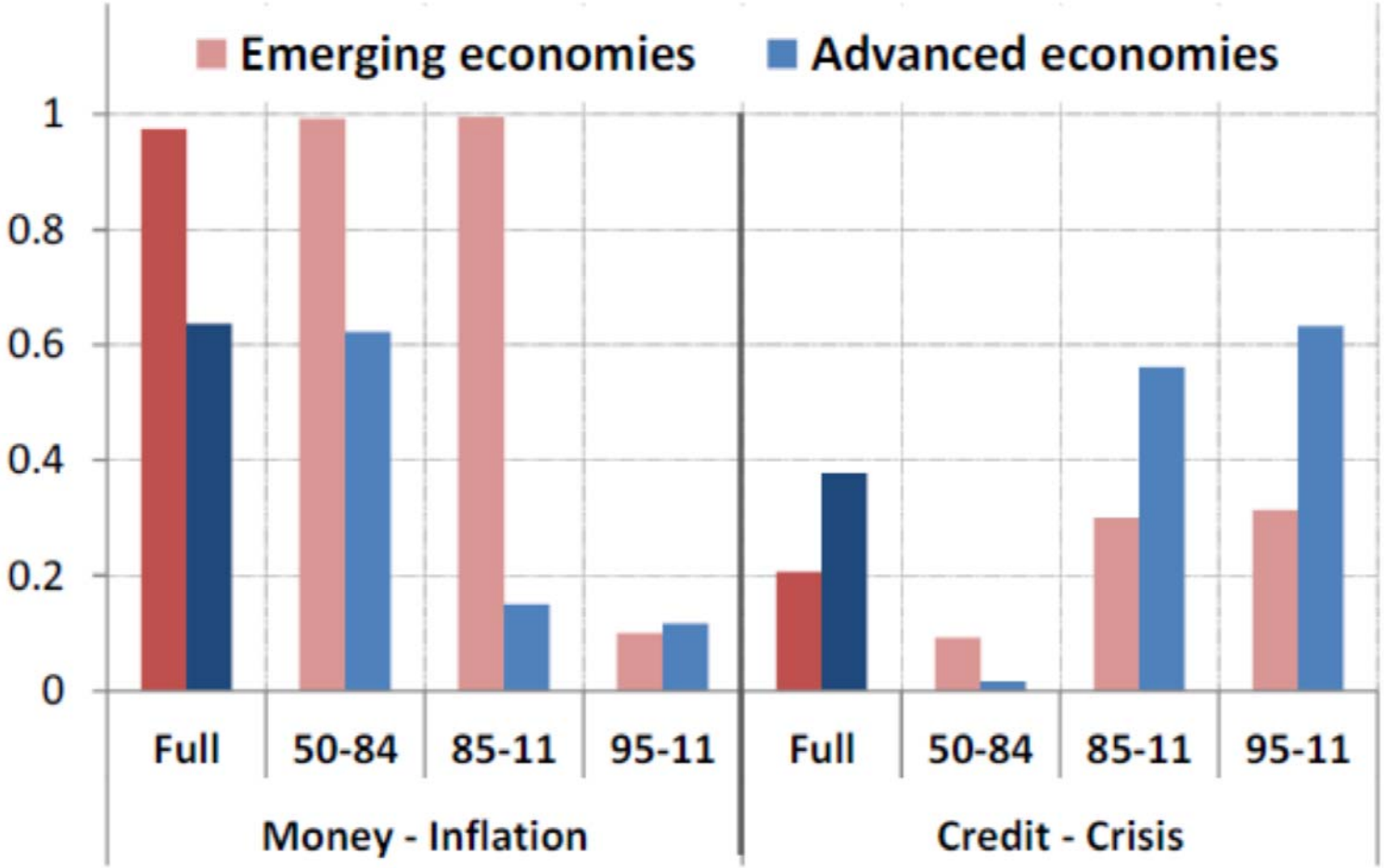
Money-inflation nexus		Credit-crisis nexus	
Long run	0.961*** (0.012)	Overall ME	0.232*** (0.050)
Error correction	-0.495*** (0.042)	AUROC	0.698*** (0.037)
Short-term dynamics		ME of individual lags	
L0	-0.192*** (0.034)	L1	0.076*** (0.024)
L1	-0.121*** (0.030)	L2	0.065*** (0.022)
L2	-0.074*** (0.020)	L3	0.074*** (0.028)
L3	-0.053** (0.022)	L4	0.014 (0.033)
L4	-0.034** (0.015)	L5	0.004 (0.026)
RMSE	0.324	Pseudo R^2	0.060
Loglikelihood	-4610.4	Pseudolikelihood	-272.2
Observations	2,336	Observations	2,331
Countries	46	Countries	46

Monetary facts over time

Money-inflation nexus				Credit-crisis nexus			
	1950-1984	1985-2011	1995-2011		1950-1984	1985-2011	1995-2011
Long run	0.797*** (0.035)	0.207*** (0.031)	0.115*** (0.018)	Overall ME	0.073 (0.055)	0.397*** (0.077)	0.415*** (0.092)
Error correction	-0.565*** (0.042)	-0.393*** (0.046)	-0.631*** (0.054)	AUROC	0.697*** (0.065)	0.729*** (0.038)	0.753*** (0.043)



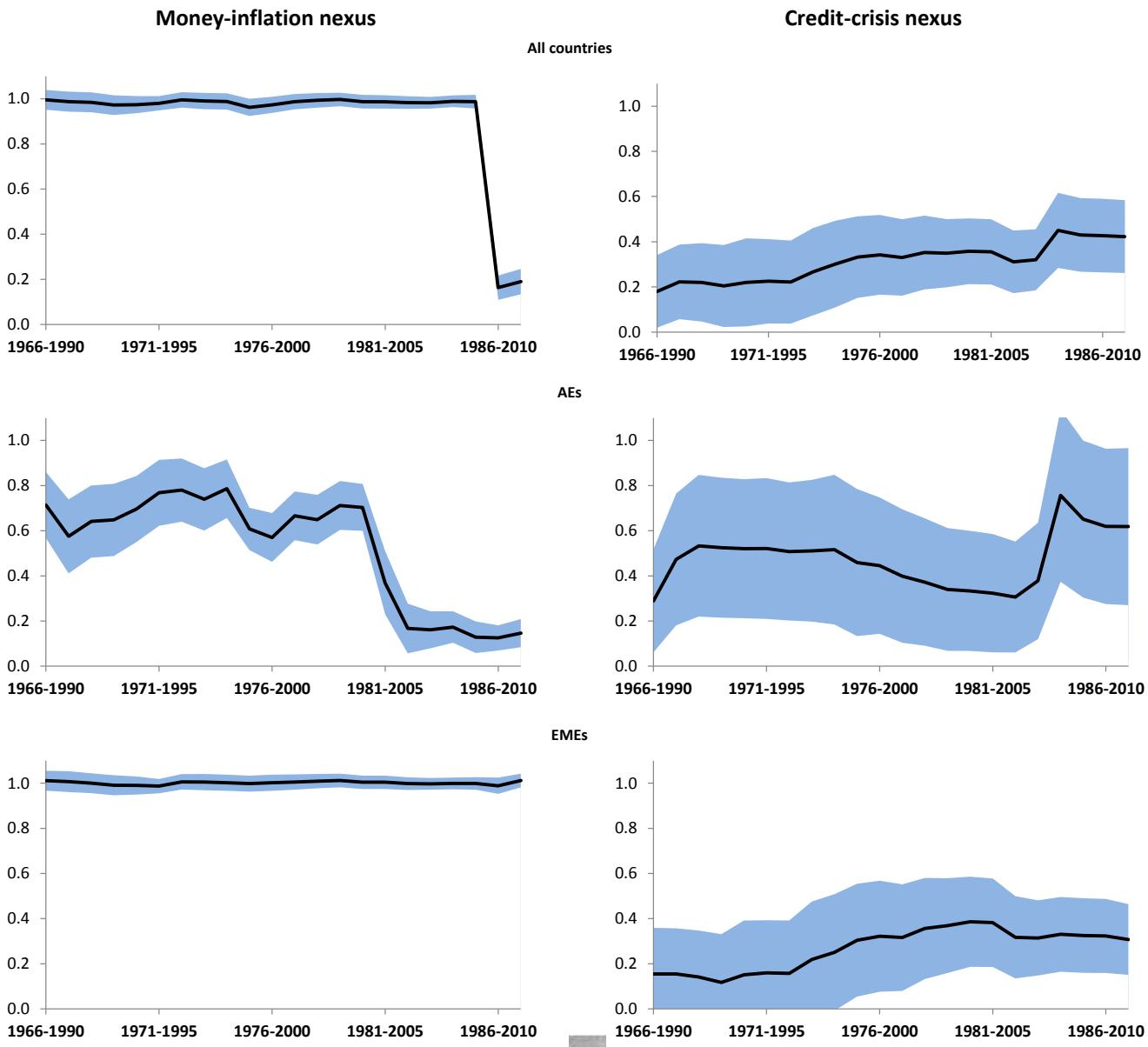
AEs vs EMEs: Sub-sample results



Robustness checks

- Wide range of robustness checks reported in the paper
- Bottomline: results are robust
- Noteworthy additional results:
  - inflation persistence and coefficients of output gap and oil price have also dropped
  - real exchange rate appreciation increases probability of financial crisis

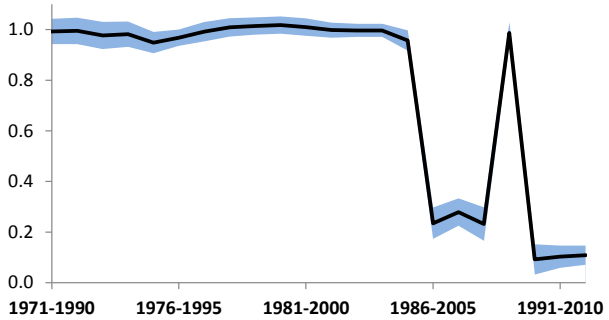
Moving window estimations (25-years)

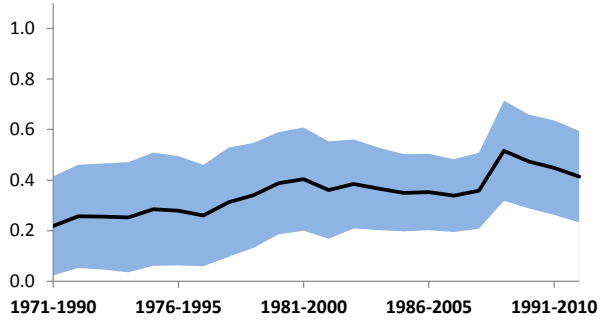


Moving window estimations (20-years)

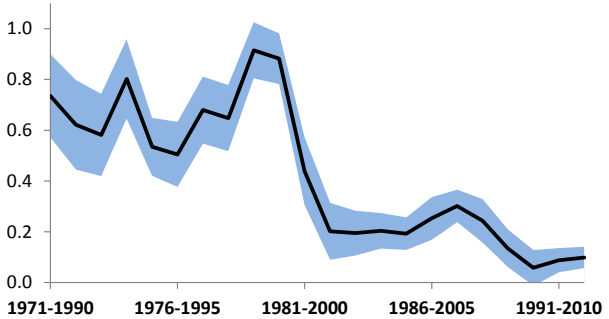
Money-inflation nexus | **Credit-crisis nexus**

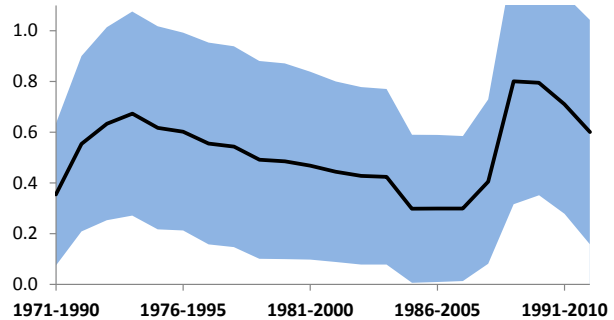
All countries



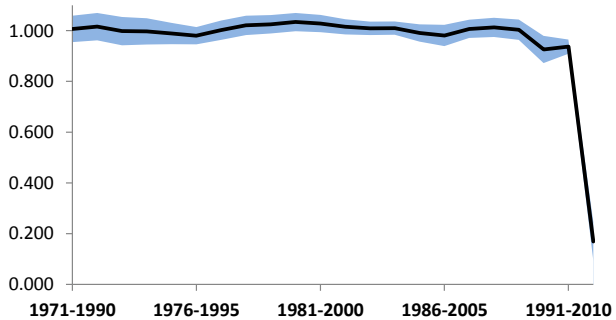


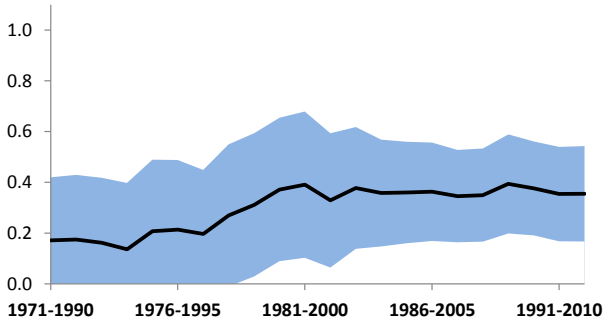
AEs





EMEs





Conclusions

- Inverse relationship between the two monetary facts
  - The money growth-inflation link is weaker in regimes characterised by low inflation and highly liberalised financial systems
  - The reverse holds true for the link between credit growth and financial crisis

- Credit-crisis link is probably stronger than indicated by the widely used estimates from historical studies (ME of 0.3 in Schularick/Taylor vs. 0.65 here for advanced economies)

Monetary policy transmission and trade-offs in the United States: Old and new

Boris Hofmann and Gert Peersman

BIS WP No 649 (July 2017)

Why could monetary transmission have changed?

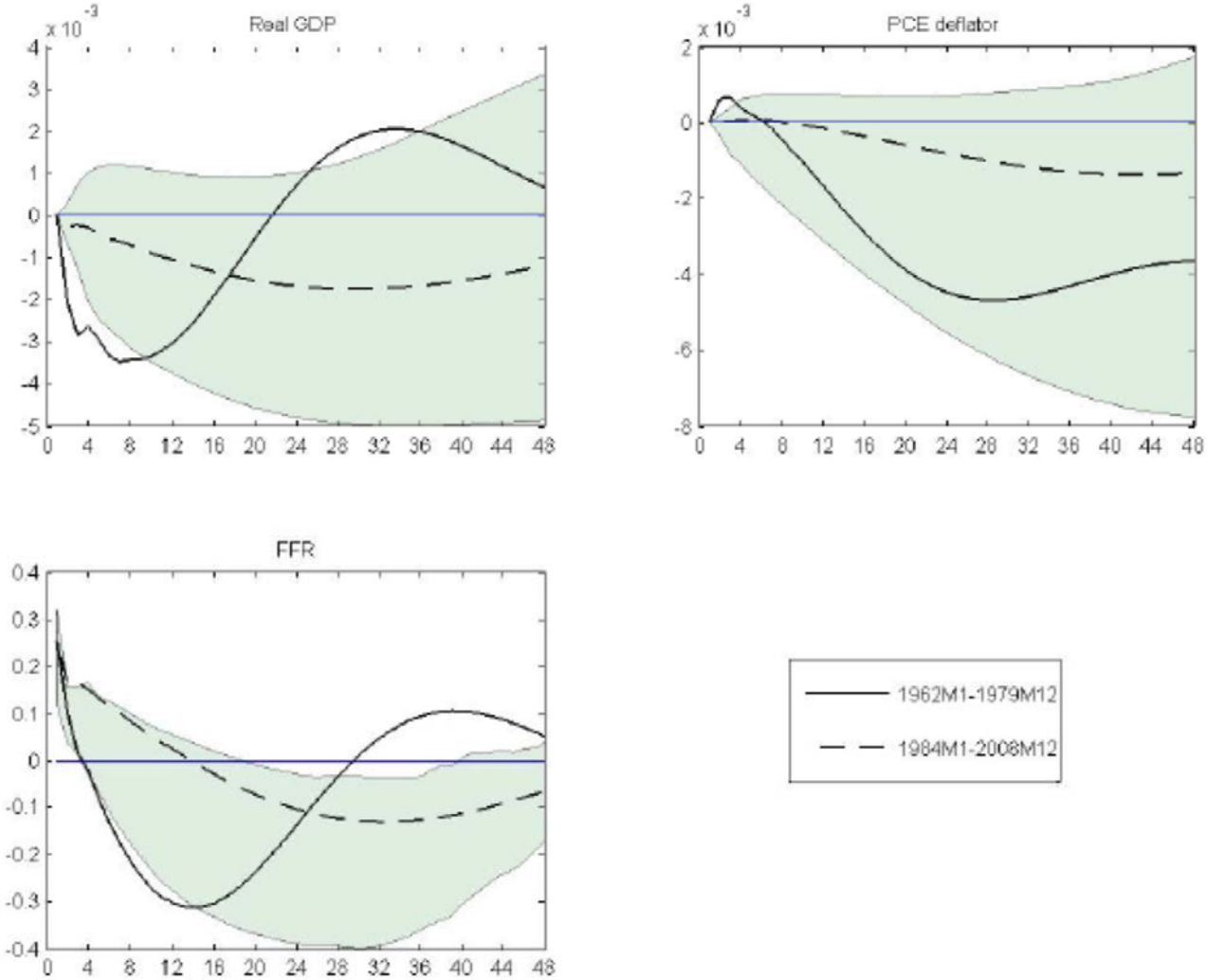
- Changes in the macroeconomy
  - Transition to low inflation regime
- Changes in financial structure
  - Abolition of Regulation Q
  - Deepening of bond markets
  - Significant changes in housing finance systems
    - Securitisation
    - Rising LTVs
    - Mortgage equity withdrawal products
- Overall implications for monetary transmission not clear *a priori*

Changes in transmission: Existing empirical evidence

- Several papers have assessed whether transmission in the U.S. has changed
  - McCarthy/Peach (2002)
  - Primiceri (2005)
  - Boivin/Kiley/Mishkin (2011)
  - Den Haan/Sterk (2011)
  - Belongia/Ireland (2015)

- Monetary transmission in the U.S. appears to have become weaker and more sluggish

Evidence from Boivin, Kiley and Mishkin (2011)



The role of house prices and credit

- Credit and in particular house prices are often not included in the empirical models

- Structural changes in the financial system suggest that house prices have become more important for monetary transmission
  - Iacoviello/Minetti (2003), Calza et al (2013): Monetary policy shock impact on house prices stronger in economies with more liberalised housing finance systems (higher LTVs)

- Against this background, U.S. monetary transmission through house prices and credit would be expected to have become stronger

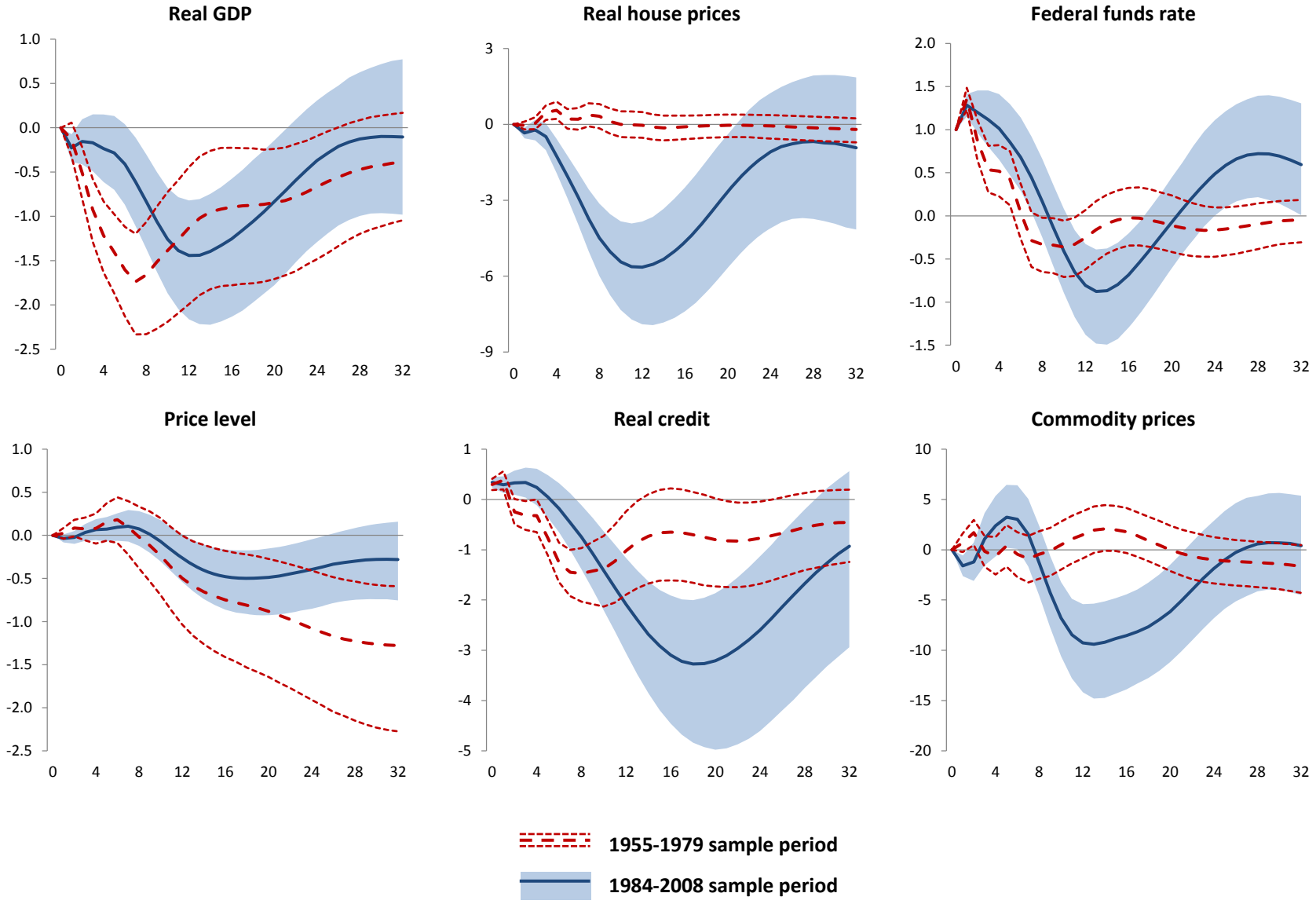
Changes in U.S. transmission: New empirical evidence

- VAR model for the U.S. economy: $Y_t = B(L)Y_{t-1} + Qw_t$
  - Y =(log commodity price index, log real GDP, log GDP deflator, log real house prices, effective FFR, log real total credit)'

- Monetary policy shock identified using standard Cholesky scheme with the above ordering

- Change over time assessed based on sub-sample estimation: 1955Q1-1979Q4 and 1984Q1-2008Q4.

Baseline VAR results



Note: Median responses to a 100 bps shock with 16th and 84th percentiles error bands; horizon is quarterly.

Peak impacts

	1955-1979		1984-2008	
	Impact	horizon	Impact	horizon
Real GDP	-1.74	7	-1.44	12
Price level	-1.28	32	-0.50	18
Real house prices	-0.20	32	-5.64	12
Real credit	-1.47	8	-3.27	18

Note: Peak effects of a 100 bps interest rate shock with corresponding horizon (quarter).

Changing trade-offs

	Shifting real GDP		Shifting the price level		Shifting real house prices		Shifting real credit	
	1955-1979	1984-2008	1955-1979	1984-2008	1955-1979	1984-2008	1955-1979	1984-2008
Real GDP	1.00	1.00	1.34	2.42	3.26	0.27	1.08	0.46
	[1.00 1.00]	[1.00 1.00]	[0.88 2.27]	[1.62 3.83]	[1.90 6.78]	[0.19 0.36]	[0.85 1.39]	[0.37 0.57]
Price level	0.75	0.41	1.00	1.00	2.43	0.11	0.81	0.19
	[0.44 1.14]	[0.26 0.62]	[1.00 1.00]	[1.00 1.00]	[1.30 4.89]	[0.07 0.16]	[0.42 1.42]	[0.11 0.32]
Real house prices	0.31	3.67	0.41	8.83	1.00	1.00	0.33	1.70
	[0.15 0.53]	[2.79 5.14]	[0.20 0.77]	[6.17 14.58]	[1.00 1.00]	[1.00 1.00]	[0.15 0.61]	[1.34 2.30]
Real credit	0.93	2.17	1.23	5.23	3.04	0.59	1.00	1.00
	[0.72 1.17]	[1.74 2.68]	[0.70 2.41]	[3.16 9.16]	[1.63 6.69]	[0.43 0.75]	[1.00 1.00]	[1.00 1.00]

Note: Estimated peak impacts on the variables in the respective row, induced by a monetary policy shock that shifts the variables in the column by 1%.
16th and 84th percentiles are in brackets.

Extended VARs: Digging deeper in the economic and financial accounts

- Additional variables enter block-exogenous VARs

$$\begin{bmatrix} Y_t \\ Z_t \end{bmatrix} = \begin{bmatrix} A(L) & 0 \\ C(L) & D(L) \end{bmatrix} \begin{bmatrix} Y_{t-1} \\ Z_{t-1} \end{bmatrix} + \begin{bmatrix} B(L) \\ E(L) \end{bmatrix} X_t + \begin{bmatrix} \mu_t^Y \\ c\mu_t^Y + \mu_t^Z \end{bmatrix}$$

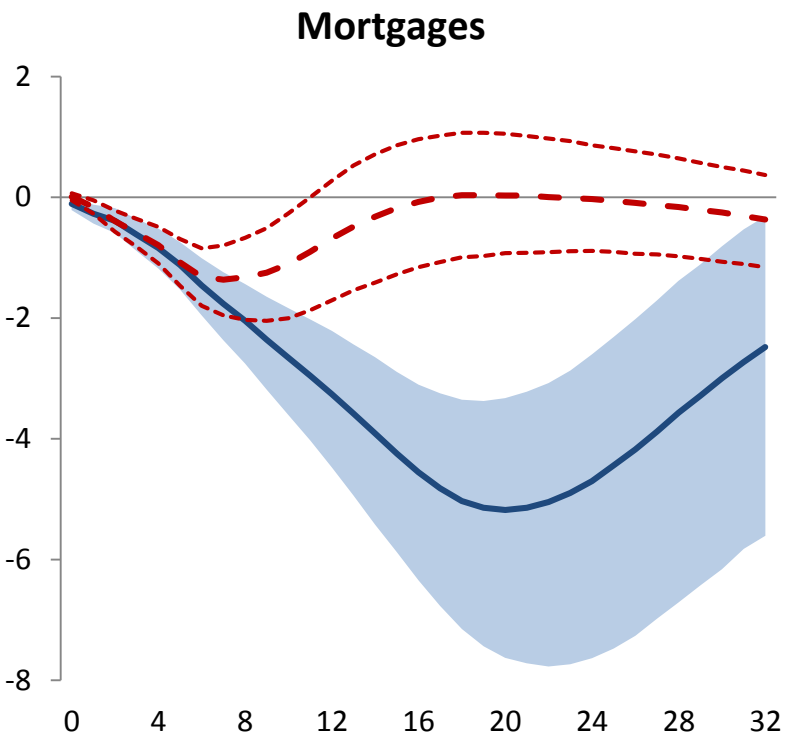
- Data from the U.S. economic and financial accounts

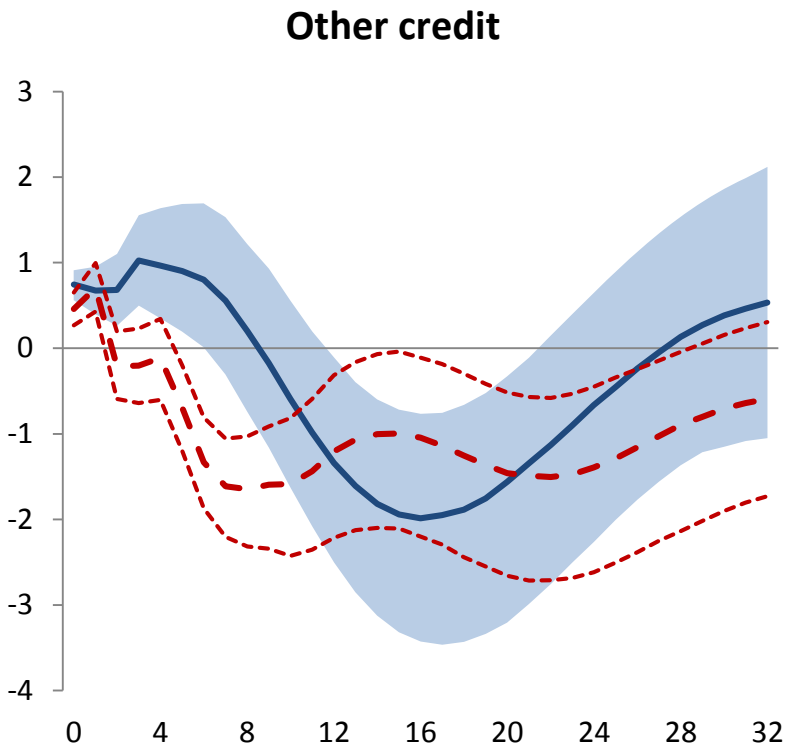
- Moving to more disaggregate level….
  - …helps to understand what is driving the aggregate results
  - …may uncover important changes not directly visible at the aggregate level

Monetary policy and credit

- What drives larger impact at the aggregate level?
- How does monetary policy impact different types of credit, different borrowers and different lenders?
- Does monetary policy get "into all the cracks"?
- Den Haan and Sterk (2010), Nelson et al (2015): It doesn't!
  - Monetary tightening reduces bank credit, but expands non-bank lending/shadow banking

Credit IRFs by instrument

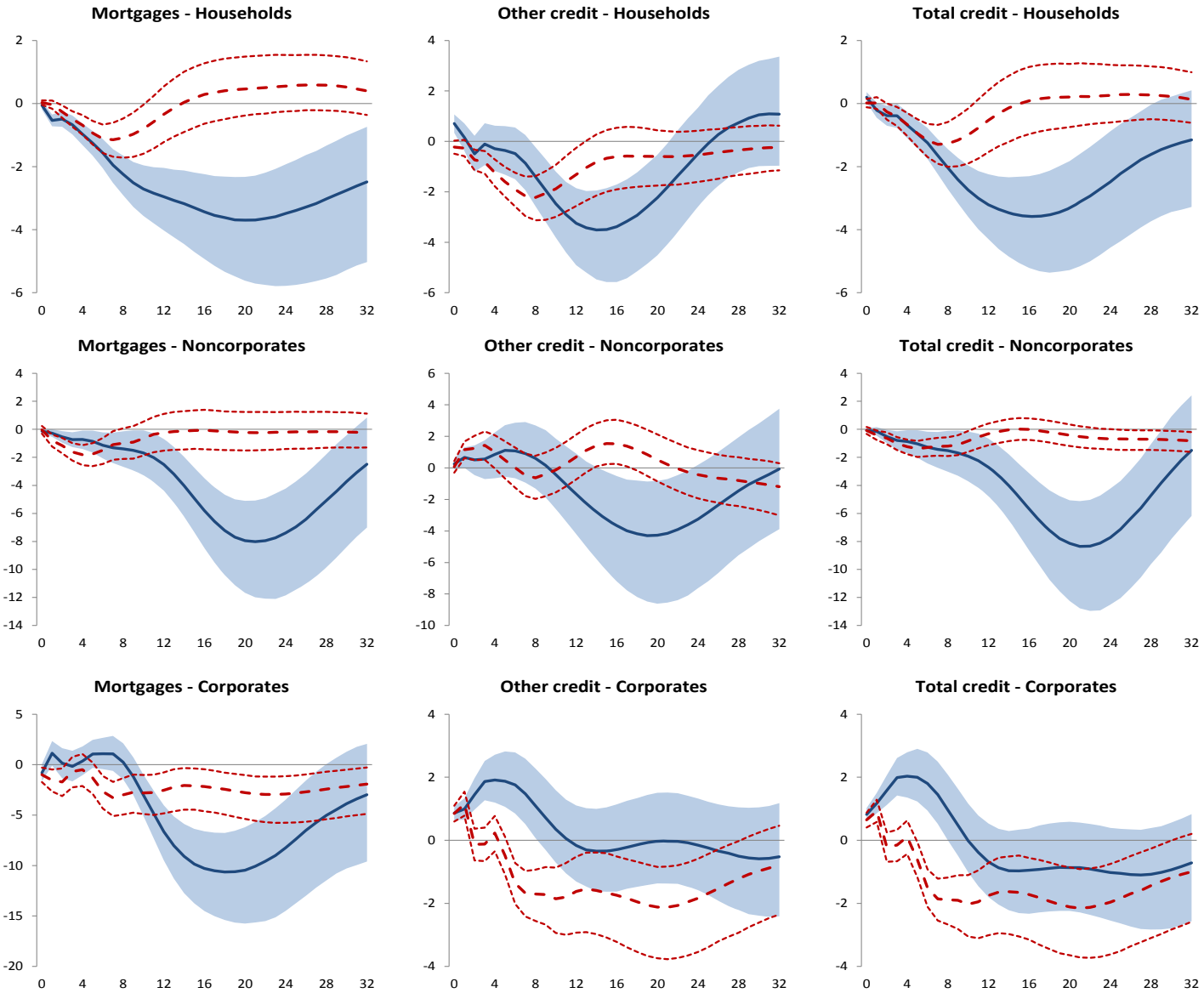




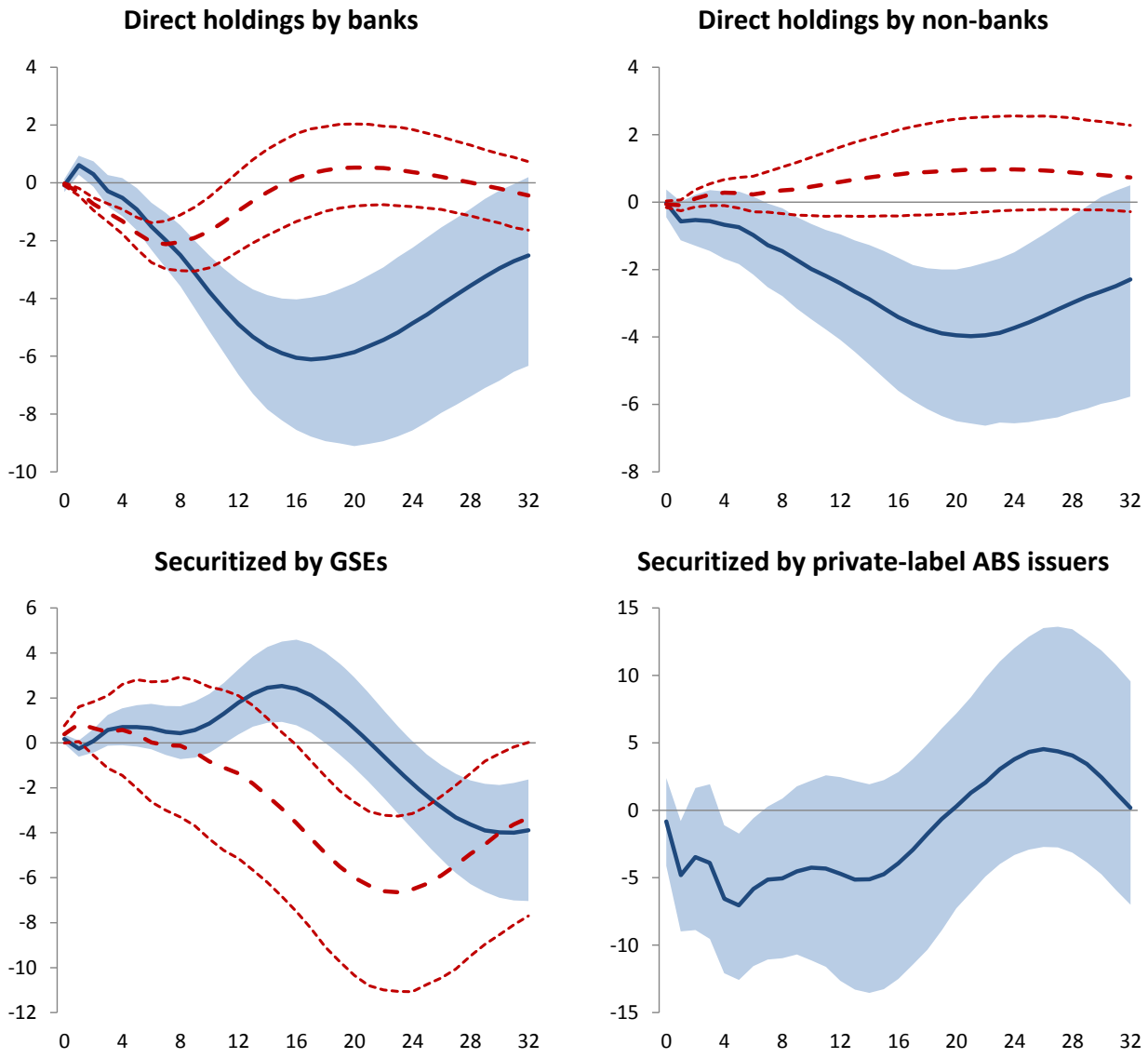
Larger impact on and larger share of mortgages

	Component/total credit (%)		Impact on component (%)		Contribution to total impact	
	1955-1979	1984-2008	1955-1979	1984-2008	1955-1979	1984-2008
Mortgages	41.1	45.8	-1.33	-5.03	-0.55	-2.31
Other credit	58.9	54.2	-1.65	-1.89	-0.97	-1.02
Total	100.0	100.0			-1.51	-3.33

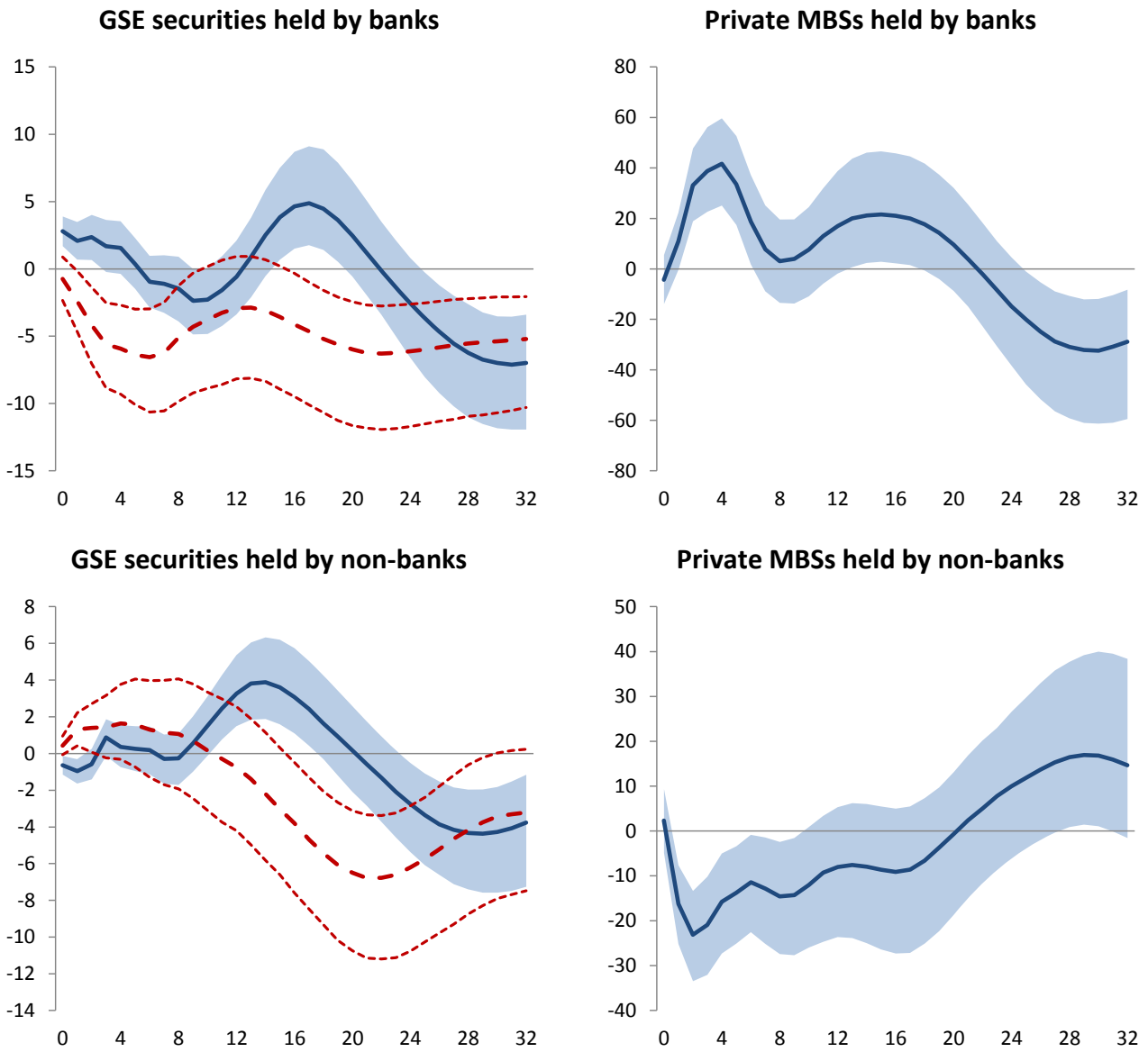
Credit IRFs by instrument and borrowing sector



IRFs of mortgage counterparties



IRFs of GSE securities and private MBSs



Conclusions

- Dynamic effects of U.S. monetary impulses on the price level have become weaker, while that on house prices and credit have considerably strengthened.

- Larger effect on total credit reflects stronger impact on and larger share of mortgages

- Mortgage lending of banks and shadow banks responds stronger to monetary impulse, GSE lending reacts counter-cyclically

- Policy implications
  - Leaning-against-the-wind may have become more effective
  - Macroeconomic stabilisation through monetary policy has probably become associated with greater financial volatility